



Delivering Near-Real-Time Insights Using AWS Lambda with Autodesk

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Up to 100X

the expected traffic supported in load testing

Reduced

the need for server and infrastructure management

Increased

developer productivity

Reduced

compute costs



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Overview

Master and city planning is complex. By using near-real-time simulations and an iterative way more informed decision-making earlier on in the process, architects and urban planners can reduce the need



for expensive and time-consuming rework later and free up more time for creative work and collaboration.

Using Amazon Web Services (AWS), engineering and design software company Autodesk built Autodesk Forma (formerly Spacemaker), a service that provides conceptual design capabilities, predictive analytics, and automations to help users make informed decisions in the earliest phase of the design and planning process. The solution is serverless and delivers fast and scalable near-real-time building simulations with reduced management and cost. The on-demand simulations accelerated near-real-time iterations by users and reduced the p90 user wait time for simulation completion.



Opportunity | Using AWS Lambda to Increase Productivity for Autodesk

Architects and design teams need near-real-time, data-driven insights to make fast, smart, and sustainable design decisions. For example, before they have invested too much time, architects want to understand the climate and temperature impacts of their design concepts. However, running the simulations to calculate these impacts can be resource intensive, and demand can be highly dependent on user traffic.

Previously, Autodesk Forma would run simulations, such as noise analysis, on a Kubernetes cluster with third-party services as task managers and message buses. (See Figure 1. Schematic illustration of the old infrastructure originally designed for Kubernetes cluster.) Given that user demand was variable, managing the cluster was a challenge. The container-based solutions scaled too slowly to meet the necessary response time.

The proper isolation of different workloads presented another challenge. One corrupted or extremely large analysis could potentially exhaust resources or bring other services down, causing disruption for users. Furthermore, infrastructure management efforts (like patching or scaling compute) took additional time, draining developer productivity and reducing the bandwidth available to deliver customer-requested features.

Additionally, solving for infrastructure management lets software teams and machine learning engineers focus more of their time on higher-impact work. The Autodesk Forma team has a set of technical core beliefs that drive the use of managed solutions so the team can focus on business logic and user interaction. A key point in the technical core beliefs is that “Our most expensive resource is hours spent by our employees. Most hours should be spent building the unique stuff only we have—our business logic and user interaction.”

The goal to improve developer productivity, scalability, and cost led the team to use [AWS Lambda](#)—a serverless, event-driven compute service that helps customers to run code—to implement a simulation engine that ran on-demand calculations to deliver insights to designers and architects.

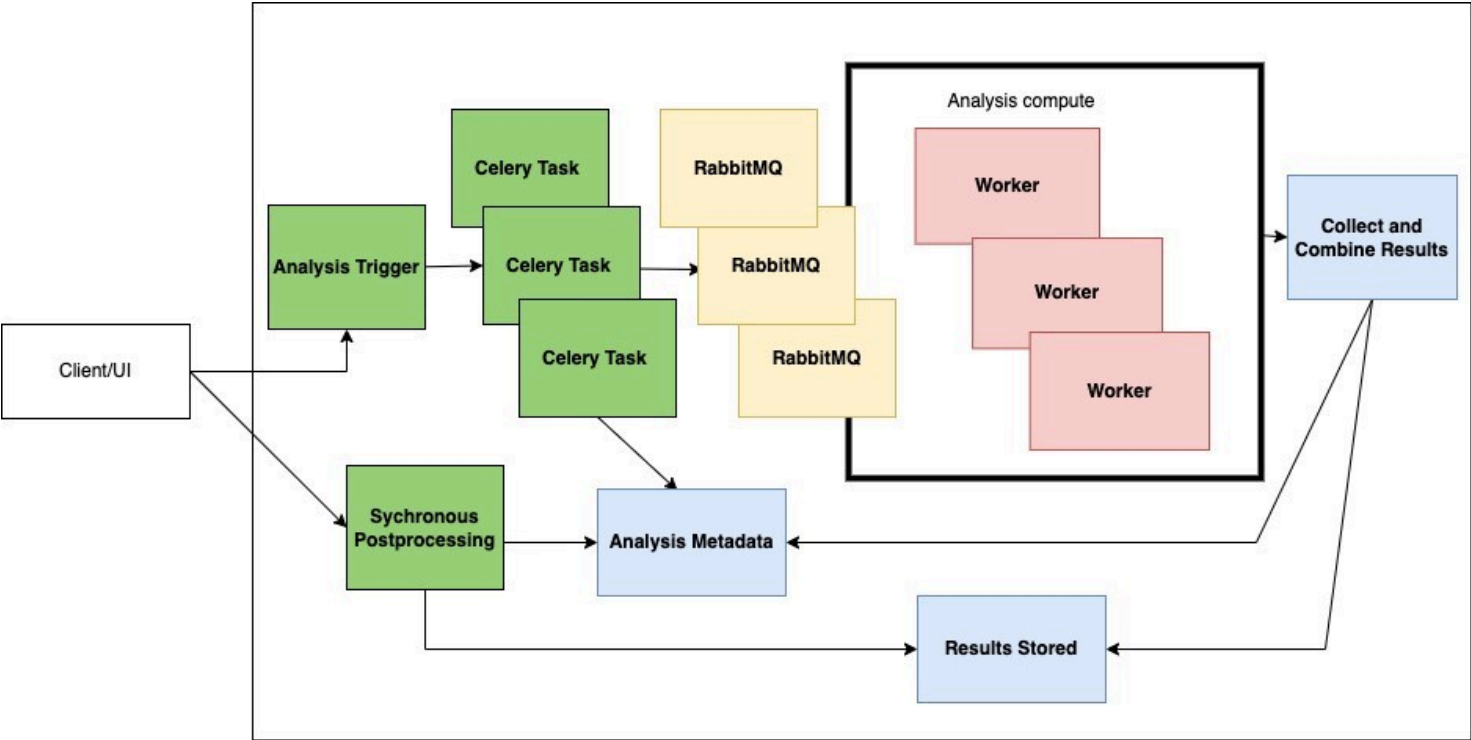


Figure 1. Schematic illustration of the old infrastructure originally designed for Kubernetes cluster.

Solution | Scaling Up to 100x the Expected Launch Traffic with AWS Lambda

The built-in, automated scaling capabilities of AWS Lambda greatly decreased simulation completion times while keeping the configuration complexity low. (See Figure 2. The current version of the serverless infrastructure used for running analyses.) The managed runtime environment reduced the engineering effort required for security patches, and the isolation between invocations improved stability.



When Autodesk Forma users make design changes, they can receive various types of support to help them make better decisions. A complicated and resource-intensive issue to solve becomes just the click of a button for the user.

When the user runs an analysis, the Autodesk Forma client submits jobs that launch a series of AWS Lambda functions. A preparation step splits the issue into smaller chunks that can be solved in parallel, using AWS Lambda’s ability to scale horizontally. When the computations are done, a collection step combines the results. Prior to launching Autodesk Forma, Autodesk worked with the AWS Lambda team on burst concurrency increases. Load tests simulating 10–100 times the expected launch traffic showed that the serverless infrastructure scales automatically without needing intervention.

To manage and orchestrate interdependent workflows while remaining serverless, Autodesk Forma added [AWS Step Functions](#), a visual workflow service that helps developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and ML pipelines. Chained and/or orchestrated lambdas in AWS Step Functions helped Autodesk Forma to go beyond simple batch computing runs. A state machine of lambdas maintains most of the AWS Lambda advantages of scalability, reduced time to production, and serverless computing.

AWS Lambda does not support GPUs, but GPU support was easy for the Autodesk Forma team to incorporate. When workflows require GPU compute—for example, when workflows use ray tracing for sunlight irradiance on a building—the Autodesk Forma team successfully combines lambdas with [Amazon Elastic Container Service](#) (Amazon ECS), a fully managed container orchestration service that simplifies the deployment, management, and scaling of containerized applications. (See Figure 3. Depiction of how the current serverless infrastructure interfaces with GPU compute instances.)

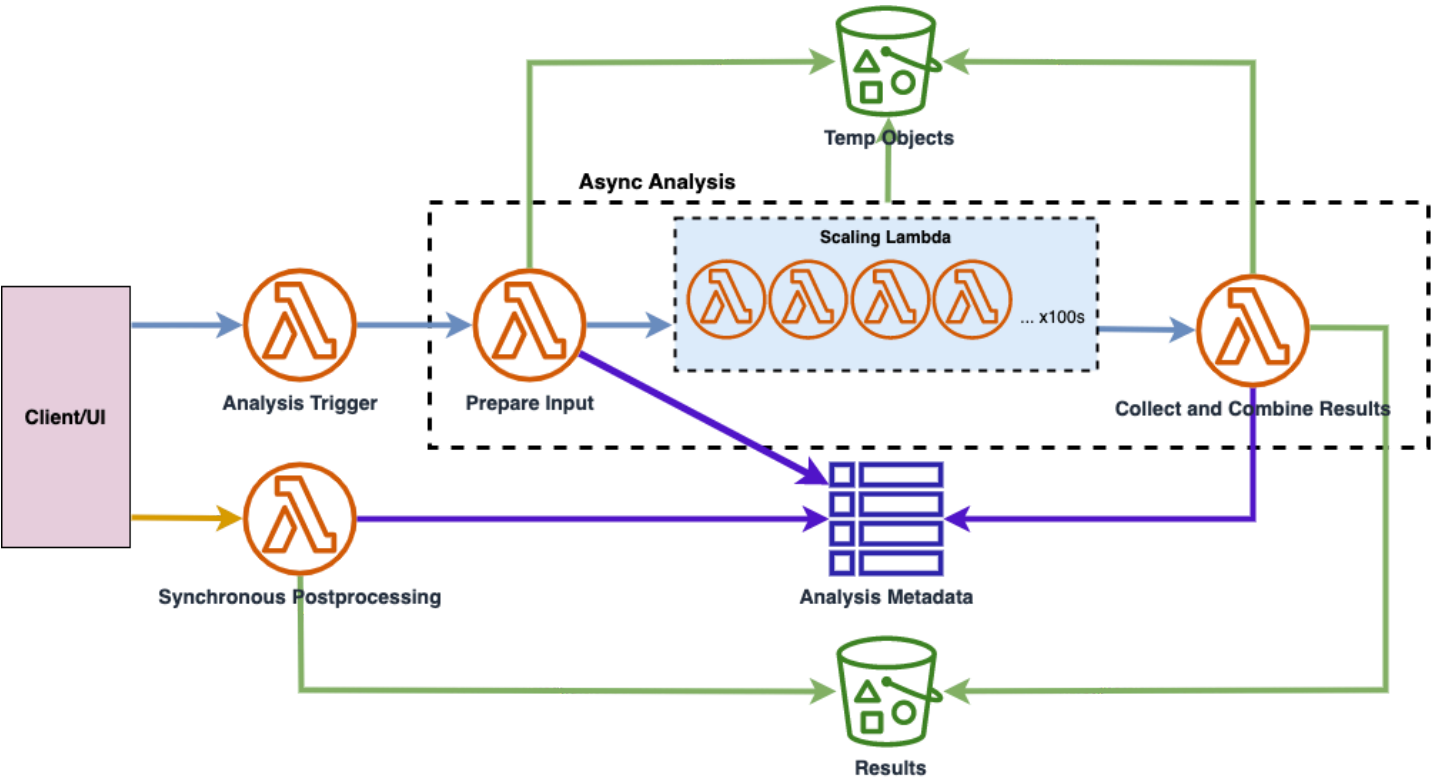


Figure 2. The current version of the serverless infrastructure used for running analyses.

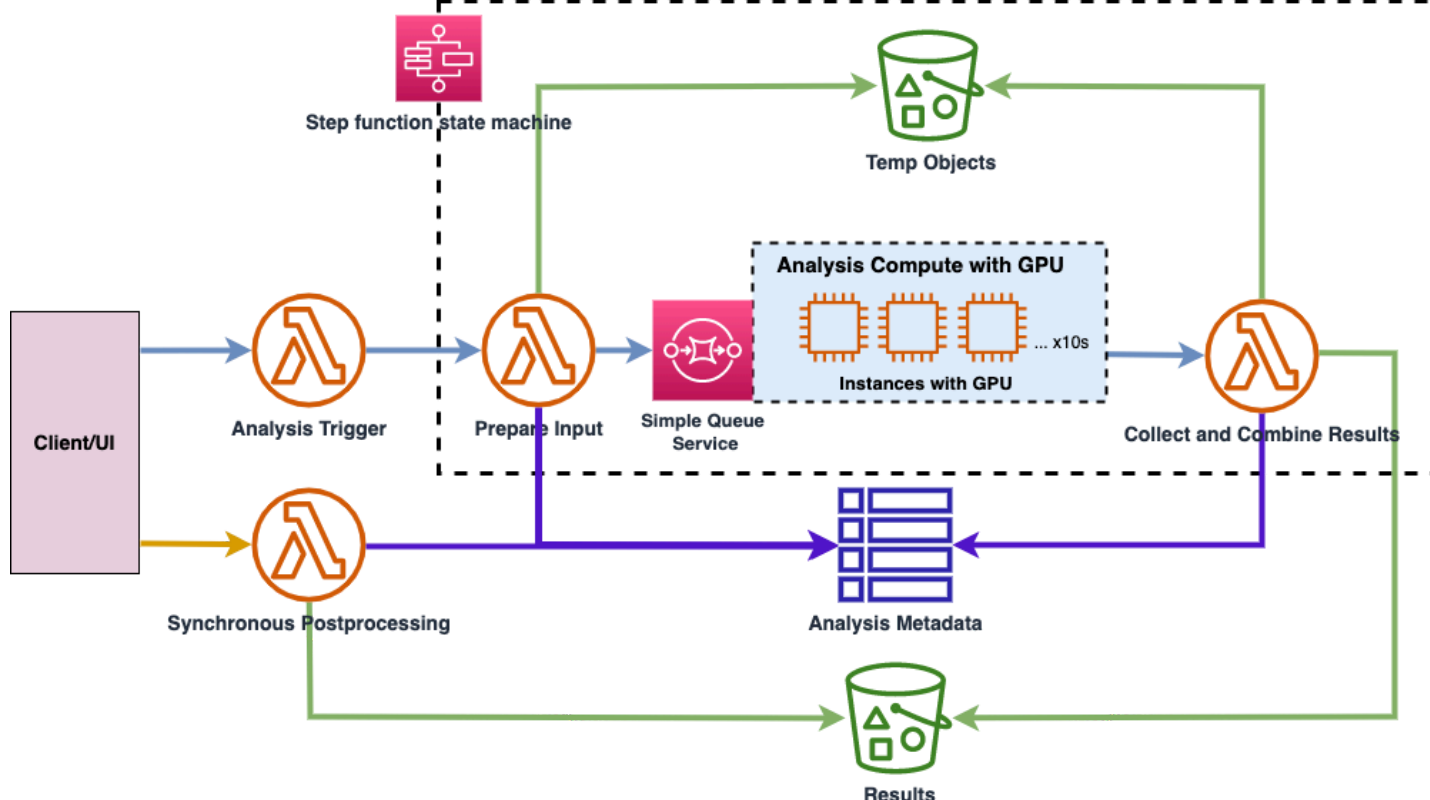


Figure 3. Depiction of how the current serverless infrastructure interfaces with GPU compute instances.

Outcome | Providing End Users with New Capabilities

The AWS architecture of Autodesk Forma has proven to be a scalable design that provides the confidence of sustaining up to 100 times the current load without any changes. Combining AWS Lambda with AWS Step Functions and Amazon ECS helps Autodesk Forma to handle end-user simulation efficiently and in near real time. The use of AWS Lambda has reduced the need for infrastructure management, increased stability of services, and reduced compute costs while reactively scaling up or down based on the current burst demand.

With AWS Lambda's ability to scale horizontally, Autodesk could free up developer resources from needing to operate and maintain a Kubernetes cluster for simulations and focus more on providing new capabilities, and a better experience, to the end users.

About Autodesk

Autodesk is a leading software provider in 3D design for the architecture, engineering, manufacturing, media, and entertainment industries. Over 100 million people worldwide use Autodesk products, which include computer-aided design and building information modeling software.

AWS Services Used

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

AWS Step Functions

AWS Step Functions is a visual workflow service that helps developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and machine learning (ML) pipelines.

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Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that simplifies your deployment, management, and scaling of containerized applications.

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